

Module 2 Cheatsheet: Use of Generative AI for Data Science

Popular GenAI tools

Name of model	Usage	Link
Hal9	EDA tool to identify key insights on data	https://www.hal9.com/
Columns.ai	Data visualization tool to create useful charts	https://columns.ai/
Akkio	Data visualization tool to create data plots like regression plots, box plots, correlation heatmaps, and so on	https://www.akkio.com/

Important prompts for generating data insights and visualizations

Task	Prompt
Generate a statistical description of data.	Write a Python code to generate the statistical description of all the features used in the data set. Include "object" data types as well.
Create regression plots between a target variable and a continuous valued source variable.	Write a Python code to generate a regression plot between a target variable and a source variable of a data frame.
Create box plots between a target and categorical source variable.	Write a Python code to generate a box plot between a target variable and a source variable of a data frame.
Evaluate parametric interdependence using correlation, p-value and pearson coefficient	Write a Python code to evaluate correlation, pearson coefficient, and p-values for all attributes of a data frame against the target attribute.
Group variables to create pivot tables. Create a p-color plot for the pivot table.	Write a Python code that performs the following actions: 1. Groups three attributes as available in a data frame df. 2. Creates a pivot table for this group, using a target attribute and aggregation function as mean. 3. Plots a pcolor plot for this pivot table.

Important prompts for model development and refinement

Task	Prompt
Linear regression between a single source attribute and target attribute and evaluate it	Write a Python code that performs the following tasks: 1. Develops and trains a linear regression model that uses one attribute of a data frame as the source variable and another as a target variable. 2. Calculates and displays the MSE and R^2 values for the trained model.
Linear regression between multiple source attributes and target attributes and evaluate it	Write a Python code that performs the following tasks: 1. Develops and trains a linear regression model that uses some attributes of a data frame as the source variables and one of the attributes as a target variable. 2. Calculates and displays the MSE and R^2 values for the trained model.
Polynomial regression model with single source and target variable	Write a Python code that performs the following tasks: 1. Develops and trains multiple polynomial regression models, with orders 2, 3, and 5, that use one attribute of a data frame as the source variable and another as a target variable. 2. Calculates and displays the MSE and R^2 values for the trained models. 3. Compares the performance of the models.
Pipeline creation for scaling, polynomial feature creation, and linear regression	Write a Python code that performs the following tasks: 1. Create a pipeline that performs parameter scaling, polynomial feature generation, and linear regression. Use the set of multiple features as before to create this pipeline. 2. Calculate and display the MSE and R^2 values for the trained model.
Grid search with ridge regression and cross validation	Write a Python code that performs the following tasks: 1. Use polynomial features for some of the attributes of a data frame. 2. Perform a grid search on a ridge regression model for a set of values of hyperparameter alpha and polynomial features as input. 3. Use cross-validation in the grid search. 4. Evaluate the resulting model's MSE and R^2 values.

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